Group Problem Set 13: Special Forces, Centripetal Acceleration and Drag

Centripetal Force is the force we experience we go around a curve in a car or on a roller coaster. We can use what we know from uniform circular motion and Newton's laws to derive the relation for it and better understand how it works.

From 2D uniform circular motion we know that acceleration is:	$a_c = \frac{v^2}{r}$
And from Newton's 2nd Law, we know:	$F = ma$
We can combine these relations to derive a relation for Centripetal Force:	$F_c = ma_c = m\frac{v^2}{r}$

Non-uniform circular motion is a related case of uniform circular motion where speed and angular velocity are not constant. There are a couple of interesting consequences to variable velocity in non-uniform circular motion:

- An object's velocity vector is always tangent to the circle.
- Speed and angular speed are not constant
- The angular acceleration is not zero.
- The net acceleration vector does not point toward the center of the circle.

We can rewrite acceleration into it's radial and tangental component:

$$a_c = a_r + a_t$$

where the radial component is responsible for the change in direction of velocity and the tangential component is responsible for the change in magnitude of velocity of an object.

1. Which way does the acceleration vector point in uniform circular motion in polar coordinates?

2. Which way does the force associated with this acceleration point (write the force in polar coordinates)?

3. Is this centripetal force a pulling or pushing force? and is it a contact or non-contact force?

4. The following problem considers a 25 kg child holding on to a a merry-go-round. The child can hold on to the merry-go-round either 0.5 m or 1 m from the center of the merry-go-round.
   (a) What is the friction required to hold a child in place as they travel 1 m/s, 0.5 m from the center of the merry-go-round?
   (b) Using this coefficient of friction, what is the max velocity the child can travel at 1 m from the center of the merry-go-round?
   (c) Did you expect the velocity to be greater or less than the velocity closer to the center of the merry-go-round?
   (d) The child's dad now wants to ride on the merry-go-round as well. How fast can the dad travel at both the 0.5 m location and 1 m location?

5. A street performer swings a 1kg flaming ball (think Maori Poi) on a 1 meter tether in a vertical circle. At a point 45^o above the plane, the ball has a speed of 5 m/s.

   (a) What is the tension on the tether?

   (b) What is the acceleration of the ball, tangential and radial components?

   (c) When the ball is straight out in front or behind the street performer (at 0 and 180^o) what is the force on the ball due to tension?

   (d) What is the velocity of the ball at each position?

   (e) When the ball is directly above and below the street performer (at 90 and 270^o), what is the force do to gravity?

   (f) Is the force due to tension the same or different than when the ball is at 0 and 180^o? Explain your answer.

   (g) What is the velocity of the ball at 90 and 270^o?